

THE SECOND BRAIN FRONTIER

Can rewiring the gut microbiome help prevent Alzheimer's disease?

FEATURE ARTICLE

NEUROSCIENCE

MICROBIOME

A literature-led exploration of the gut–brain axis, early evidence, and the role of digital health.

By Kenisya Saravenen | BSc Biomedical Sciences

5 min read · #AlzheimersDisease #GutBrainAxis #Microbiome #DigitalHealth



Why look beyond the brain?

A long-standing neurological problem is opening a new biological front.

For decades, Alzheimer's research has concentrated on amyloid-beta, tau pathology and inflammation within the central nervous system. The gut–brain hypothesis asks whether systemic signals and the microbiome could also influence the environment in which neurodegeneration develops.

THE CORE QUESTION

If gut microbial changes are linked to cognitive decline, can changing the gut actually change the course of disease?

Evidence currently points to an interesting association and plausible mechanisms — but causality in humans remains unresolved.

THE DEBATE

Brain-first view

Alzheimer's is primarily driven by pathological processes in the brain — including amyloid, tau and neuroinflammatory signalling.

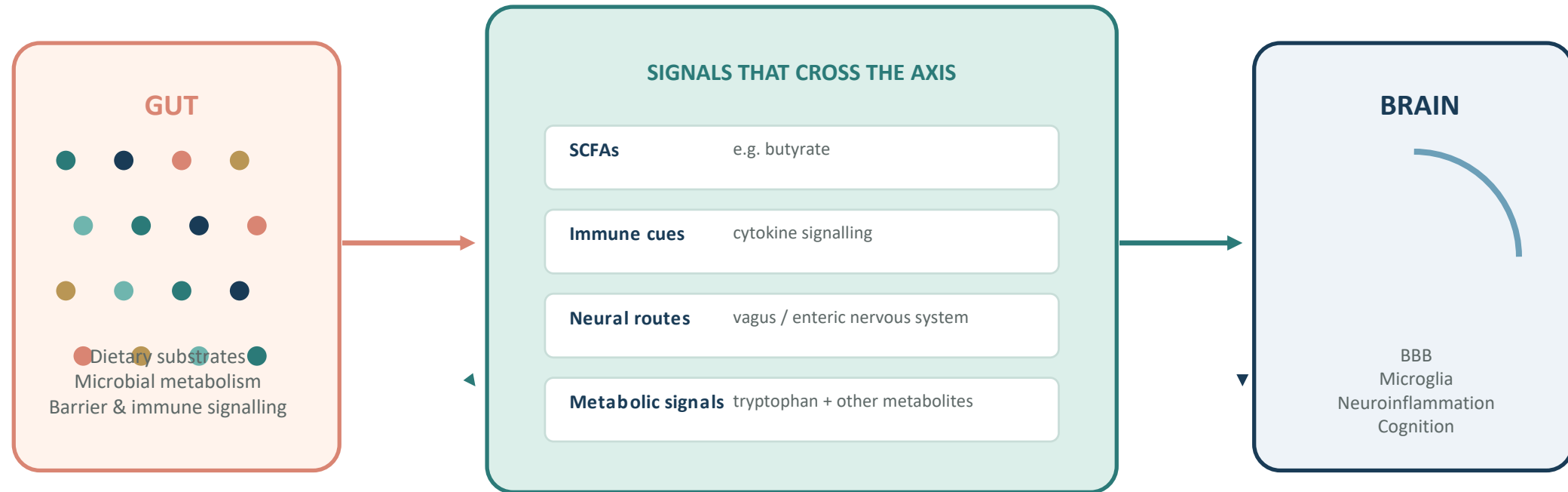
Gut–brain view

The gut microbiome, microbial metabolites and immune signalling may modify neuroinflammation and disease susceptibility.

Neither view alone explains the entire evidence base.

The gut–brain axis is a biological network, not a single pathway

Microbes communicate with the host through metabolites, immune signalling and neural routes.



Two-way communication

The point is not that “bad bacteria cause Alzheimer’s”, but that the microbiome is one component of a complex signalling network.

What does the human evidence actually show?

The signal is compelling enough to study — not strong enough to call causal.

37

human studies reviewed

A 2026 systematic review of microbiome studies published in 2021–2025 found broadly consistent dysbiosis patterns in Alzheimer’s disease.

ASSOCIATION ≠ CAUSATION

Heterogeneity across populations, methods and taxa limits direct comparison.

Microbiome composition

Reduced diversity and changes in several short-chain-fatty-acid-producing taxa have been reported in Alzheimer’s cohorts.

Human observational evidence

Intervention signal

A 2026 review identified 15 intervention studies involving 4,275 participants; improvements were reported especially in prodromal or mild impairment.

Promising — but heterogeneous

What the evidence cannot tell us yet

- Whether dysbiosis contributes to cognitive decline or partly reflects disease, ageing, diet or medication.
- Which specific microbial changes are beneficial, harmful, or simply biomarkers of a broader physiological shift.
- Whether microbiome-targeted intervention can prevent or delay dementia in large, long-term human trials.

Sources: Libriani et al. 2026 (15 intervention studies); 2026 systematic review of 37 human AD microbiome studies.

From association to mechanism: the most striking evidence is still preclinical

Animal models suggest the microbiome can modify neuroinflammatory and amyloid-related phenotypes.

SCIENCE · 2023

ApoE × microbiota × tauopathy

Seo et al. used a genetically engineered mouse model of tauopathy. Germ-free or antibiotic-perturbed conditions reduced gliosis, tau pathology and neurodegeneration in an ApoE-dependent manner.

What it adds

Evidence that the microbiota can influence neurodegenerative phenotypes — but in mice, not humans.

PMID 36634180 · Science 379(6628):eadd1236

MOL PSYCH · 2026

Butyrate and gut–brain amyloid

Brossaud et al. reported that early butyrate supplementation reduced gut and brain amyloid pathology, neuroinflammation and memory deficits in an Alzheimer’s mouse model.

What it adds

A biologically plausible metabolite with effects across gut and brain — but again, preclinical evidence only.

Mol Psychiatry 31:3910–3928 · 2026

TAKE-HOME

The strongest mechanistic findings currently sit one translational step away from the clinic.

Sources: Seo et al., Science (2023); Brossaud et al., Molecular Psychiatry (2026).

Prevention already looks multidomain

The microbiome hypothesis fits into a wider evidence base for lifestyle-based risk reduction.

FINGER · 2015

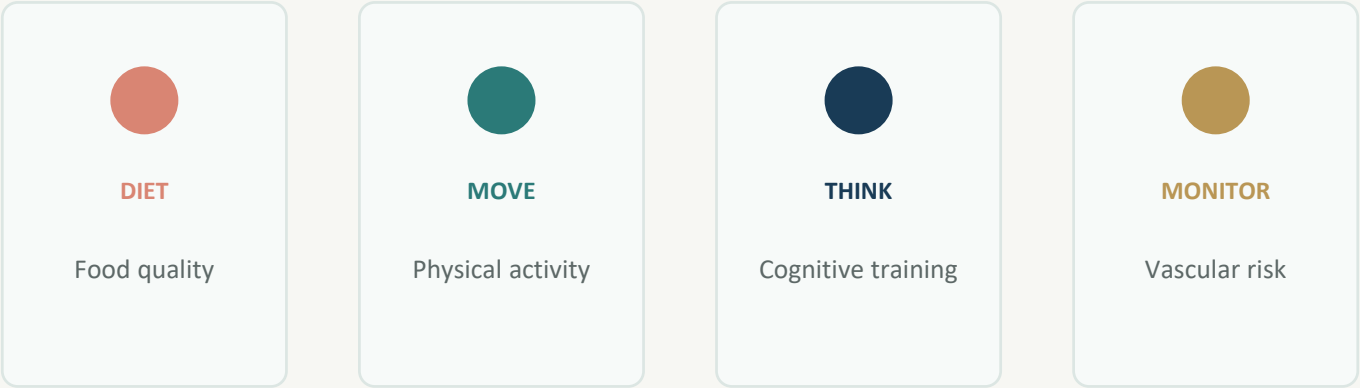
1,260 participants

60–77 years
Elevated dementia risk

2-year multidomain intervention

Diet + exercise + cognitive training +
vascular-risk monitoring

Improved overall cognitive performance compared with general health advice.



What this means for the gut–brain hypothesis

A microbiome-targeted strategy is more likely to be clinically useful as one component of a broader prevention framework than as a stand-alone “cure”.

The evidence base for lifestyle prevention is currently stronger than the evidence base for microbiome-specific prevention.

Source: Ngandu et al., The Lancet (2015), PMID 25771249. FINGER was not a microbiome trial.

Where could AI meal scanning fit?

A proposed tool for behaviour change — not a proven Alzheimer's therapy.



What is realistic today?

AI meal-photo systems can analyse food images and return dietary feedback. A 2026 pilot trial protocol, FiberMore, is testing AI-powered meal-photo logging to support higher dietary fibre intake.

PROPOSED DIGITAL CONCEPT

What is not proven?

There is currently no evidence that an AI nutrition app can prevent Alzheimer's disease. Its value would need to be demonstrated through behaviour, biomarker and — ultimately — cognitive outcomes.

Source: FiberMore AI-based meal-photo logging trial protocol (2025/2026); this slide distinguishes tested technology from the proposed Alzheimer's application.

So, can we “rewire” the gut?

The scientifically defensible answer is: promising, plausible — but not yet proven.

What we know	What remains uncertain
Gut microbiota differ in many people with cognitive impairment / Alzheimer’s cohorts.	Whether dysbiosis is a cause, consequence, or co-traveller of disease.
Microbial metabolites can influence immune, barrier and neural signalling.	Which taxa or metabolites are actually causal in humans.
Mouse studies show microbiome-dependent effects on tau, amyloid, inflammation and memory.	Whether these mechanisms translate into meaningful human clinical benefit.
Lifestyle interventions can protect cognitive performance in at-risk older adults.	Whether microbiome-targeted interventions add benefit beyond established prevention strategies.
THE NEXT EXPERIMENT — Large, longitudinal randomised trials linking microbiome changes → biomarkers → cognition.	

Evidence synthesis based on human reviews, FINGER, and preclinical microbiome studies cited in the article.

The future of brain health may be less brain-only than we once thought.

The gut microbiome is emerging as a plausible modifier of neuroinflammation, barrier function and disease biology. Yet the field is still crossing the gap between association, mechanism and clinical proof.

PROMISING BIOLOGY

OPEN QUESTIONS

EARLY INTERVENTION

For now, the strongest case is not for a miracle microbiome “cure”, but for better research into how everyday biology — diet, microbes, immunity and lifestyle — interacts with healthy brain ageing.

Selected references

Vancouver-style references used in the feature.

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2. Seo DO, O’Donnell D, Jain N, et al. ApoE isoform- and microbiota-dependent progression of neurodegeneration in a mouse model of tauopathy. *Science.* 2023;379(6628):eadd1236. doi:10.1126/science.add1236.
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4. Libriani S, Facchinetti G, Marti F, et al. The association between gut microbiota and cognitive decline: a systematic review of the literature. *Nutr Res.* 2026;147:16–31. doi:10.1016/j.nutres.2026.01.003.
5. Microbiome alterations in Alzheimer’s disease: a systematic review of current evidence and global perspectives. 2026. PMID:41929953. doi:10.1177/25424823261436287.
6. Brossaud R, Oullier T, Bessard A, et al. The short-chain fatty acid butyrate prevents gut-brain amyloid- β pathology and neuroinflammation in an Alzheimer mouse model. *Mol Psychiatry.* 2026;31:3910–3928. doi:10.1038/s41380-026-03522-6.
7. FiberMore: AI-based mHealth intervention to increase dietary fibre intake using meal-photo logging. Trial protocol. PMID:41343777.

Evidence note: claims about Alzheimer’s prevention should be read in light of the distinction between observational human evidence, intervention trials, and animal models.